NETWORKING · NET-06

BreakFixLearn

VLANs & 802.1Q Trunking

Use this when: you split a switch into separate networks, carry them between switches on a trunk, or route between them.

CCNA 200-301 · topics 2.1, 2.2
Sheet v1.0 · Oct 2026

Scan to visit

The 802.1Q tag: 4 bytes added on a trunk

IEEE 802.1Q

Dest MAC	Source MAC	TPID 0x8100	PCP	DEI	VLAN ID	Type	Payload	FCS
6 bytes	6 bytes	16 bits: "tagged"	3: priority	1	12 bits: 1 to 4094	2	data	4

Access port: one VLAN, no tag **Trunk:** many VLANs, each tagged **Native VLAN:** the one sent untagged (default 1)

VLAN numbers

1	Default; can't delete. Don't use for users.
2 to 1001	Normal range, your VLANs
1002 to 1005	Reserved (Token Ring, FDDI)
1006 to 4094	Extended range

Does a trunk form? (DTP)

Default: dynamic auto

	AUTO	DESIRABLE	TRUNK	ACCESS
auto	No	Yes	Yes	No
desirable	Yes	Yes	Yes	No
trunk	Yes	Yes	Yes	Mismatch

Configure

Access port · trunk port

```
vlan 10
 name SALES
interface g0/1
 switchport mode access
 switchport access vlan 10
 switchport voice vlan 150  ! IP phone
```

```
interface g0/24
 switchport mode trunk
 switchport nonegotiate     ! no DTP
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,150
 switchport trunk allowed vlan add 30
```

Route between VLANs

```
! router on a stick
interface g0/0.10
 encapsulation dot1q 10
 ip address 192.0.2.1 255.255.255.192
```

```
! layer 3 switch (SVI)
ip routing
interface vlan 10
 ip address 192.0.2.1 255.255.255.192
 no shutdown
```

`show vlan brief`	VLANs and access ports
`show interfaces trunk`	Trunks, native, allowed
`show interfaces g0/1 switchport`	Mode of one port

Stop VLAN hopping

DANGER Ports left in dynamic mode let an attacker negotiate a trunk. Set `access` or `trunk` plus `nonegotiate`.

CAUTION Use an unused native VLAN (999) and the same one on both ends.

CAUTION `allowed vlan 30` **replaces** the list. Use `add`.

Trunk down? Check

Native VLAN	Same on both ends? CDP logs a mismatch
Allowed list	Is the VLAN on both trunks?
VLAN exists	Created on the far switch too?

REMEMBER

- One VLAN = one subnet
- **auto + auto** = no trunk
- Shut unused ports, park them in an unused VLAN

Sources: IEEE 802.1Q · Cisco Catalyst VLAN and VLAN trunk configuration guides · CCNA 200-301 exam topics
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